



PRIMARY SCIENCE

TOPICAL REVISION

SOIL



SUMMARIZED SCIENCE REVISION NOTES

SOIL

Definition; Soil is the top layer of the earth where plants grow.

SOIL COMPOSITION

It is made up of: -

- Mineral particles
- Living things
- Humus
- Water
- Air

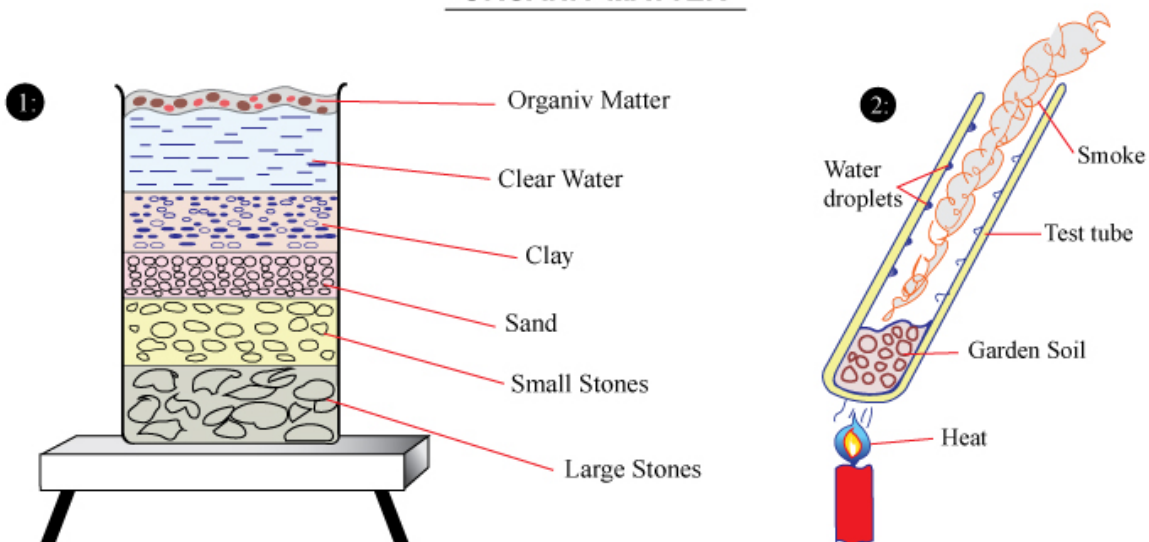
1. Humus

- Add water and shake
- Let it settle water and soil settle

Organic matter

Organic matter refers to the dead remains of both plants and animals. Organic matter can also be called humus.

ORGANIC MATTER



Observation

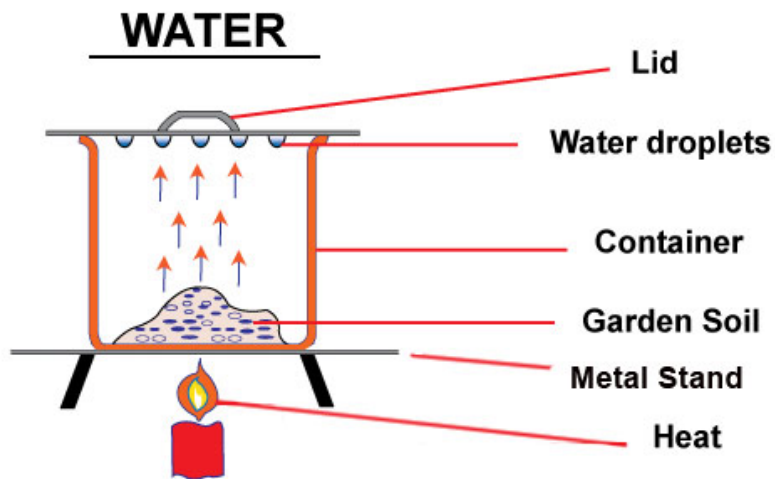
In diagram 2 smoke that smells like burning rubbish is seen.

This shows the soil contains organic matter.

2. Water

Heat garden soil in a beaker or a test tube.

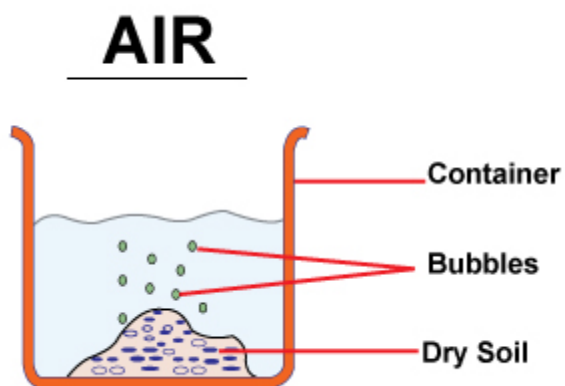
Cover the beaker with a glass plate.



Water droplets are formed on the glass plate and on the walls of the test tube.

3. Air

Pour water in a beaker containing dry garden soil



- Bubbles of air are seen escaping



4. Living things

- Put fresh garden soil in a muslin bag.
- After a few days, water will rise in the glass.
- Put fresh garden soil in a muslin bag.
- Lime water will turn milky after some time.
- Living organism takes in oxygen and gives out carbon dioxide which turns lime water milk.

TYPES OF SOIL

There are three types

- a) Clay soil
- b) Sand soil and
- c) Loam soil
 - Soil texture is the roughness or smoothness of soil
 - Rough soil has larger particles .e.g. sand
 - Smooth soil has fine or small particles .e.g. clay

SOIL DRAINAGE

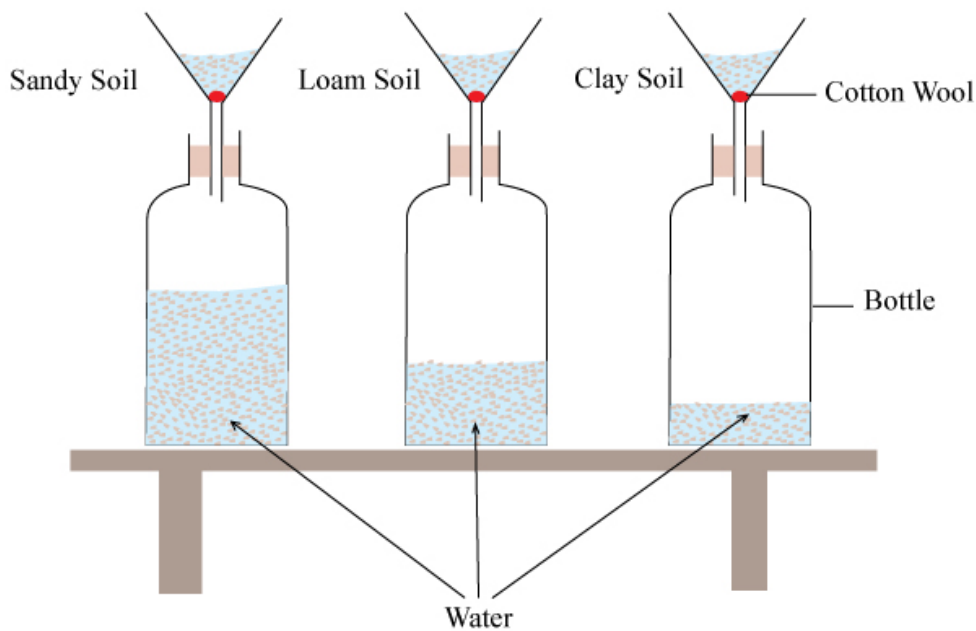
- This is the flowing of water through the soil particles.
- The rate of flow depends on the size of soil particles.
- Large particles allow water to pass through easily while small particles do not allow water to pass easily.

To show drainage

Have three different types of soil: Sand, Clay, Loam

Amount of

- a) Soil in each funnel must be equal
- b) Water to be used for draining must be equal
- c) Time taken must be the same in all soils
- d) Cotton wool in each funnel be equal



Observations

- The first drops of water were observed through sandy soil and the last in clay soil.
- Greatest amount of water was collected under sandy soil, then loam and least in clay soil.

Conclusion

- Sandy soil has the highest drainage.
- Clay soil has the lowest drainage.
- Loam soil has moderate drainage.

WATER RETENTION

- This is the water-holding power in different soils.
- Water retention is determined by the soil particles.
- Clay soil has smallest particles and highest water-holding power.
- Sandy soil has the largest particles and lowest water-holding power.
- Loam soil has moderate water retention.

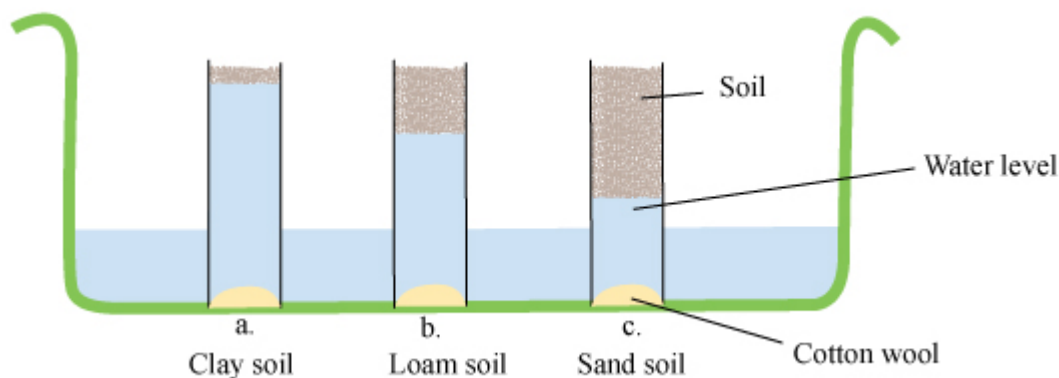
SOIL CAPILLARITY

This is the rising of water in the soil.

Material needed:-

- Dry sand, clay and loam soils of equal amounts
- Equal amounts of cotton wool or cloth.
It helps in holding soil particles in the tubes
- Clear tubes of same size (diameter)

- Experiment must take equal time in all tubes.
- Amount of water does not matter.



Observation

- Water rises highest in clay soil.
- Water rises lowest in sandy soil.
- Water rises moderately in loam soil.

Conclusion

- Clay soil has the highest capillarity.
- Sandy soil has the lowest capillarity.
- Loam soil has moderate capillarity.

Note: - Capillarity is determined by the size of the soil particles.

- Clay soil has the finest particles and highest capillarity.
- Sand has the largest particles and lowest capillarity.
- Loam soil has particles and moderate drainage.

CHARACTERISTICS OF DIFFERENT SOILS

a) Clay

- Forest particles
- Small air space
- Highest capillarity
- Lowest drainage
- Cracks when dry
- It is a heavy soil
- Expands when wet
- Sticky when wet and used in modeling.
- Does not spread easily

**b) Sandy**

- Has large air space
- Has rough texture
- Drains water faster
- Spreads easily when dry
- It is a light soil
- Has lowest capillarity
- Has poorest water retention
- Used in building

c) Loam

- Has average size of soil particles
- Average texture
- Good for farming
- Moderate capillarity
- Moderate drainage

SOIL EROSION

It is the removal and carrying away of the top fertile soil by agents of soil erosion.

Agents of Soil Erosion

They help to remove and carry away soil particles. Examples:

- Water
- Wind

a) Wind

Moving air carries away loose top fertile soil.

b) Water

Rain and flowing water carry loosely packed particles on bare ground.

Factors Affecting Soil Erosion

- a) Animals - They loosen soil particles which are easily carried by water and wind.
- b) Amounts of rainfall – Heavy rainfall increases the rate of erosion.
- c) Slopes – Soil erosion is higher because soil is not firmly held.
- d) Vegetation cover - erosion is higher on an area with no vegetation.
- e) Type of soil - Loose soil is easily carried away
- f) Human activities - This includes poor farming methods e.g. deforestation, cultivating along the slopes, burning vegetation cover, overgrazing, construction of roads, mining.

Types of Soil Erosion

There are four types

- a) Splash erosion (spatter)
- b) Sheet erosion
- c) Rill erosion
- d) Gulley erosion

a) Splash Erosion (Spatter)



- It is caused by raindrops hitting the soil with force forming holes.

- It occurs in ground with loose soil.

b) Sheet Erosion



It is the uniform carrying away of soil on a flat bare surface or gently sloping land.

It is not easily noticed

Landslide is a form of sheet erosion

Wind is the main agent of soil erosion.

c) Rill Erosion



Running water leads to formation of small rills.

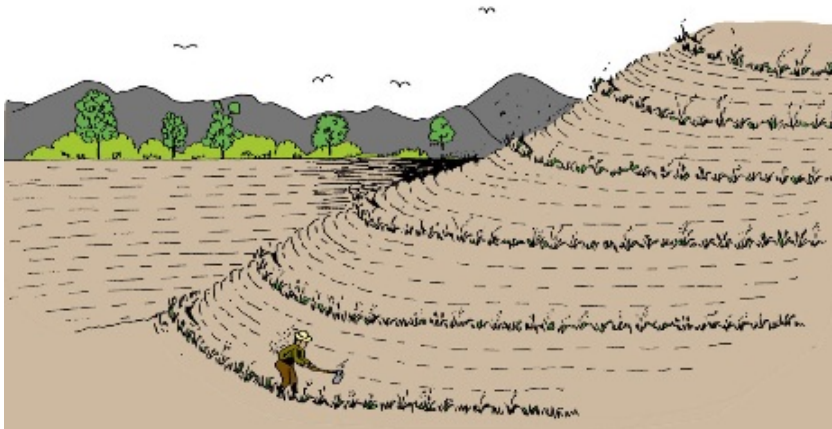
Gulley Erosion

Rills continue to widen and deepen to form V-shaped and U-shaped channels called gullies.



How to Prevent Soil Erosion

- Contour ploughing or cultivating across the slope



- Planting trees and cover to hold soil particles
- Mulching
- Crop rotation
- Construction of cut-off drains
 - Proper stocking rate
- Avoid burning of vegetation
- Construction of terraces and gabions to check gulley erosion



SOIL FERTILITY

It is the ability of the soil to produce and sustain high crop yield.

Soil can lose fertility through

- Poor farming methods
- Soil erosion
- Minerals absorbed by plants

Fertilizers



- They are organic and inorganic substances added to the soil to improve fertility.
- They introduce or increase the quantity of minerals in the soil.

Manures

They are made of organic materials that can easily decompose or not.

Examples:-

- Green manure
- Farmyard manure
- Compost manure

Green Manure

- Growing plants are ploughed or dug into the soil while still green.
- Ploughed plants rot to enrich soil with nutrients
- Such plants should have qualities like

- High growth rate
- High decomposition rate
- Leafy
- Leguminous e.g. groundnuts, beans, peas, are best

Farmyard Manure

- It is made from animal waste .e.g. dung, urine and animals' beddings.
- Goats, cows, sheep, poultry are used.
- Bedding includes grass, maize straws, sawdust.
- It should be protected from sunlight and rainfall to conserve nutrients.

Compost Manure



- It is made from plants and other organic waste e.g. kitchen waste, banana and potato peelings.
- The material is piled in pits or in a heap.
- The heaped material decomposes to form compost manure.
- Regular turning is done to facilitate even decay.



Wood ash helps to add nutrients .e.g. potassium

Water keeps material moist but not wet

Manure injects or adds useful bacteria that help in decomposition.

Soil introduces decomposing bacteria

N.B Plastics, polythene bags/bottles do not rot and should not be put in the heap.

Artificial or Inorganic Fertilizers

They are manufactured in the factories.

They are named according to the nutrients they carry.

There are two groups:

- a) Straight fertilizers
- b) Compound fertilizers

Straight Fertilizers

- They carry one major or macro-nutrient..
- Macro-nutrients are required by plants in large quantities .e.g. nitrogen, phosphorus and potassium.

There are three types of straight fertilizers:

Nitrogenous fertilizers which carry nitrogen;

Phosphatic fertilizers carry phosphorus;

Potassic fertilizers carry potassium.

Compound Fertilizers

They contain more than one major nutrient such as Diammonium phosphate D.A.P.

Advantages of Manures and Fertilizers



- They increase growth rate
- They increase the amount of produce as well as the quality
- They improve soil fertility
- Manures do not pollute the soil
- Manures last longer in the soil
- Inorganic fertilizers supply nub specific nutrients to the plants
- Chemical fertilizers are easy to transport.

Disadvantages of Manures and Fertilizers

- Nutrients in organic manures are mixed
- Chemical fertilizers can pollute soil if used excessively.
- Chemical fertilizers can burn crops.
- Green manure is expensive to make.
- Manures are bulky and require labour.
- Some chemical fertilizers are corrosive.
- Some chemical fertilizers, when exposed to sunlight, lose nutrients.